Theory and Experiments on Self-Starting Siphons

A One-Dimensional Slug-Flow Model with Layered Analysis

IYPT 2026 — Problem #1

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Abstract

A self-starting siphon primes spontaneously when partially immersed, requiring no external suction. We develop a one-dimensional slug-flow model through systematic layered analysis, progressively incorporating physical effects to understand how parameters control priming success and filling time.

Starting from the Navier-Stokes equations, we derive the governing ordinary differential equation through explicit simplifications: incompressibility, axisymmetry, quasi-steady velocity profiles, and rigid-slug dynamics. The frictionless limit yields the energy integral $I(L)$, computable analytically for piecewise-linear geometries. Container drawdown introduces the area ratio A/A_0, while friction adds pipe roughness ε/D and bend losses K. Comparing these three analytical integrals reveals the energy cost of each effect.

Experimental validation using video tracking demonstrates that the full numerical model captures the essential dynamics, while the analytical bounds provide physical insight unavailable from simulation alone. Validity limits—cavitation, capillarity, entrance effects, and secondary flows—are quantified through dimensionless parameters (Bond, capillary, Dean, and Womersley numbers) and visualized in regime diagrams.

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1 Introduction

The International Young Physicists' Tournament (IYPT) 2026 Problem #1 poses the following challenge:

> *"A self-starting siphon can be made using a piece of rigid tubing bent into a specific shape. When the siphon is partially immersed in water, it begins siphoning water without the need for initial suction. Investigate how the relevant parameters, such as the geometry, affect the siphoning process."*

1.1 Physical Background

The siphon has fascinated scientists since antiquity. While its operation was long attributed to atmospheric pressure, modern understanding recognizes that gravity provides the fundamental driving mechanism (Boatwright et al., 2015). In a standard siphon, liquid flows from a higher reservoir to a lower one through an inverted U-tube, even though part of the path rises above the source reservoir. The key insight is that the liquid column acts as a continuous chain: water descending the outlet leg "pulls" water up the inlet leg through cohesive forces.

A *self-starting* siphon eliminates the need for manual priming. Through clever geometry—typically involving an internal drop and rise sequence—the siphon uses momentum gained during an initial descent to carry water over a crest above the waterline. Once primed, ordinary siphon action drains the container.

1.2 Previous Work

The physics of siphons has been revisited periodically in the literature. McDonald (2024) provides a detailed force analysis, while Potter & Barnes (1971) explores the role of molecular cohesion. The Bernoulli equation approach is standard in fluid mechanics textbooks (White, 2011; Fox et al., 2011), though the transient filling dynamics of self-starting siphons have received less attention.

1.3 Approach and Contributions

This paper develops a theoretical framework through layered analysis:

1. **Navier-Stokes derivation** (section 3): We begin from first principles, making each simplification explicit and showing how the governing equation transforms at each step.
2. **Frictionless model** (section 4): With friction neglected, geometry alone determines the outcome through the energy integral $I(L)$.
3. **Drawdown effects** (section 5): The area ratio A/A_0 controls how the container level falls.
4. **Friction effects** (section 6): Pipe roughness and bend losses are incorporated using the Darcy-Weisbach equation (Brown, 2002).
5. **Model comparison** (section 7): Three analytical integrals visualize the hierarchy of effects.
6. **Numerical dynamics** (section 8): Velocity-dependent friction requires numerical integration.
7. **Experimental validation** (section 9): Video tracking provides velocity measurements.
8. **Validity limits** (section 10): Dimensionless parameters quantify when assumptions fail.

2 Physical Setup

2.1 Coordinate System and Geometry

We work in a vertical plane with horizontal coordinate x (rightward positive) and vertical coordinate h (upward positive), with datum $h = 0$ at the container base.

The pipe centreline connects five nodes by straight segments:

$$A = (x_0, h_0), \quad B = (x_1, h_1), \quad C = (x_2, h_2), \quad D = (x_3, h_3), \quad E = (x_4, h_4). \tag{1}$$

Each node has a physical role:

Node	Physical Role
A	Inlet, near container bottom
B	First crest, below initial water level
C	Low point of internal drop (velocity peaks here)
D	Main crest, *above* the initial water level
E	Outlet, at or below container bottom

The defining constraint for self-starting is $h_1 < h_c^{(0)} < h_3$: the first crest lies below the waterline, the main crest above it.

2.2 Arclength Parameterization

The arclength L measures distance along the pipe from the inlet. On segment i (for $L \in [S_{i-1}, S_i]$), the centreline elevation follows:

$$h(L) = c_i + s_i L, \tag{2}$$

where $s_i = \sin\theta_i = (h_i - h_{i-1})/L_i$ is the slope and $c_i = h_{i-1} - s_i S_{i-1}$ ensures continuity.

2.3 Critical Points

Point W marks where the descending front crosses the waterline for the last time on segment $D \to E$:

$$h(S_W) = h_c^{(0)} \quad \Rightarrow \quad S_W = \frac{h_c^{(0)} - c_4}{s_4}. \tag{3}$$

Point W' marks the ascending crossing on segment $C \to D$. Between W' and W, the front is above the waterline and experiences negative driving force.

2.4 Experimental Parameters

Table 1 lists all parameters for the worked example, derived from our experimental setup.

Table 1: Complete parameter set for the experimental siphon.

Geometry			
Node	x (cm)	h (cm)	Role
A	0.0	1.0	Inlet
B	7.5	11.0	First crest
C	13.5	5.0	Low point
D	21.4	13.0	Main crest
E	30.3	-1.7	Outlet
Initial water level: $h_c^{(0)} = 12.5\,\text{cm}$		Total length: $L_{\max} = 49.41\,\text{cm}$	
Pipe Properties			
Inner diameter	$D = 5\,\text{mm}$	Cross-section	$A = 0.196\,\text{cm}^2$
Relative roughness	$\varepsilon/D \approx 3.5 \times 10^{-4}$	Minor losses	$\sum K \approx 14.4$ (fitted)
System Parameters			
Container area	$A_0 = 340\,\text{cm}^2$	Area ratio	$A/A_0 = 5.77 \times 10^{-4}$
Water density	$\rho = 998\,\text{kg/m}^3$	Viscosity	$\nu = 1.0 \times 10^{-6}\,\text{m}^2/\text{s}$
Surface tension	$\gamma = 0.073\,\text{N/m}$	Gravity	$g = 9.81\,\text{m/s}^2$

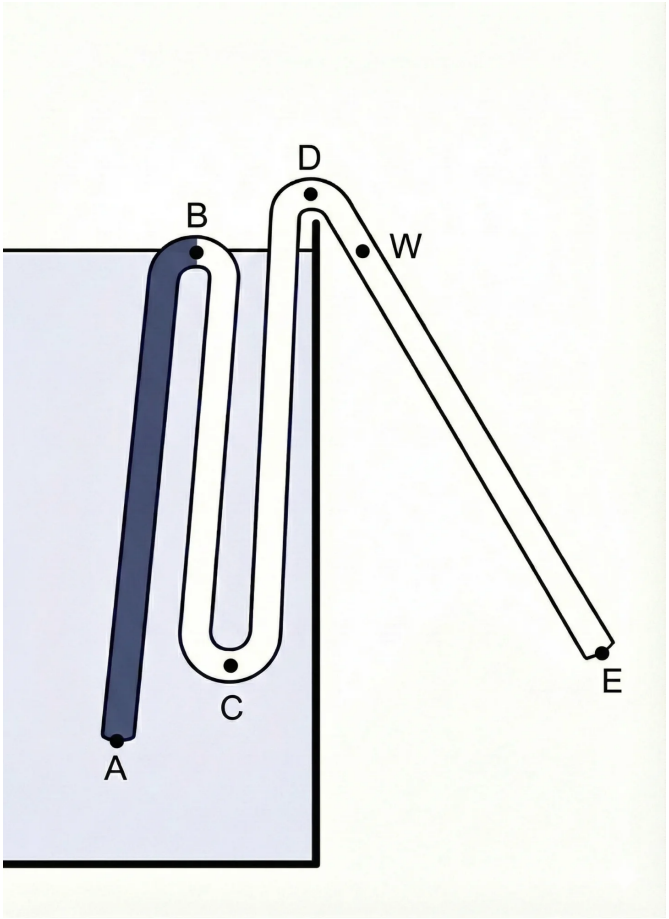


Figure 1: Schematic of the self-starting siphon. The water level $h_c^{(0)}$ lies between crests B and D. Segment $A \to B$ fills hydrostatically; the drop $B \to C$ builds momentum; the climb $C \to D$ passes above the waterline; point W marks the final waterline crossing.

3 Derivation from Navier-Stokes

We derive the governing equation systematically from the Navier-Stokes equations, making each simplification explicit. This approach clarifies what the model captures and what it neglects.

3.1 The Full Equations

For an incompressible Newtonian fluid, conservation of mass and momentum yield (Batchelor, 1967):

$$\nabla \cdot \mathbf{u} = 0, \tag{4}$$

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u}\right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}. \tag{5}$$

These partial differential equations describe velocity $\mathbf{u}(\mathbf{x}, t)$ and pressure $p(\mathbf{x}, t)$ at every point in space and time.

3.2 Step-by-Step Reduction

We now apply a sequence of assumptions, showing how each transforms the equations.

Assumption 1 (Axisymmetric pipe flow)**.** *The pipe has circular cross-section with diameter D. In cylindrical coordinates (r, θ, s) aligned with the pipe axis, velocity is $\mathbf{u} = u_r \hat{r} + u_s \hat{s}$.*

Under this assumption, the continuity equation (4) in cylindrical coordinates becomes:

$$\frac{1}{r}\frac{\partial (r u_r)}{\partial r} + \frac{\partial u_s}{\partial s} = 0. \tag{6}$$

The axial momentum equation (5) reduces to:

$$\rho\left(\frac{\partial u_s}{\partial t} + u_r \frac{\partial u_s}{\partial r} + u_s \frac{\partial u_s}{\partial s}\right) = -\frac{\partial p}{\partial s} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_s}{\partial r}\right) + \frac{\partial^2 u_s}{\partial s^2}\right] + \rho g_s, \tag{7}$$

where $g_s = -g \sin\theta$ is the gravitational component along the pipe.

What this neglects: Three-dimensional effects at bends, non-circular cross-sections.

Assumption 2 (Slender pipe: $D \ll L$)**.** *Radial velocities are much smaller than axial velocities: $u_r \ll u_s$. Axial gradients are much smaller than radial gradients: $\partial/\partial s \ll \partial/\partial r$.*

Applying this assumption to (7):

$$\rho\left(\frac{\partial u_s}{\partial t} + u_s \frac{\partial u_s}{\partial s}\right) = -\frac{\partial p}{\partial s} + \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_s}{\partial r}\right) + \rho g_s. \tag{8}$$

The radial diffusion term $\mu r^{-1}\partial_r(r\partial_r u_s)$ describes how viscosity redistributes momentum across the pipe cross-section.

What this neglects: Very short pipes where entrance effects dominate.

Assumption 3 (Quasi-steady velocity profile)**.** *The instantaneous velocity profile matches the steady-state profile for the current mean velocity.*

For laminar flow, this is the parabolic Hagen-Poiseuille profile (Hagen, 1839; Poiseuille, 1840):

$$u_s(r, s, t) = 2\bar{u}(s, t)\left(1 - \frac{4r^2}{D^2}\right), \tag{9}$$

where $\bar{u}$ is the cross-sectional mean velocity.

This assumption is valid when flow changes slowly compared to the viscous diffusion time:

$$\tau_\nu = \frac{D^2}{\nu}. \tag{10}$$

For $D = 5\,\text{mm}$ and water, $\tau_\nu \approx 25\,\text{s}$. Since filling occurs in $\sim 2\,\text{s}$, this assumption is marginal during rapid acceleration.

What this neglects: Profile development (entrance length), unsteady friction enhancement.

Assumption 4 (Cross-sectional averaging)**.** *We integrate over the pipe cross-section, replacing point values with area-averaged quantities.*

Integrating the axial momentum equation (8) over the cross-section $\mathcal{A}$:

$$\int_{\mathcal{A}} \rho \frac{\partial u_s}{\partial t}\, dA + \int_{\mathcal{A}} \rho u_s \frac{\partial u_s}{\partial s}\, dA = -\int_{\mathcal{A}} \frac{\partial p}{\partial s}\, dA + \int_{\mathcal{A}} \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_s}{\partial r}\right) dA + \int_{\mathcal{A}} \rho g_s\, dA. \tag{11}$$

The viscous term integrates to the wall shear stress. Using the Darcy-Weisbach formulation (Brown, 2002):

$$\int_{\mathcal{A}} \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_s}{\partial r}\right) dA = -\tau_w \pi D = -\frac{f\rho\bar{u}^2}{8}\pi D = -\frac{f\rho\bar{u}^2}{2D}A, \tag{12}$$

where f is the Darcy friction factor. This yields:

$$\rho A\frac{\partial \bar{u}}{\partial t} + \beta\rho A\bar{u}\frac{\partial \bar{u}}{\partial s} = -A\frac{\partial \bar{p}}{\partial s} - \frac{f\rho\bar{u}^2}{2D}A + \rho g_s A, \tag{13}$$

where β is the momentum flux correction factor ($\beta = 4/3$ for laminar flow, $\beta \approx 1$ for turbulent).

What this neglects: Radial pressure variations, detailed velocity profile shape.

Assumption 5 (Rigid slug with uniform velocity)**.** *The entire liquid column moves as a rigid body with uniform velocity $v(t)$ along its length.*

This eliminates spatial gradients of velocity ($\partial\bar{u}/\partial s = 0$) and allows treating the slug as a lumped mass. The momentum equation becomes:

$$\rho AL\frac{dv}{dt} = -\Delta p \cdot A - \frac{f\rho v^2}{2D}LA + \rho g A\Delta h, \tag{14}$$

where L is the slug length, Δp is the pressure difference between inlet and front, and Δh is the elevation difference.

What this neglects: Acoustic waves, compressibility, spatial velocity variations.

Assumption 6 (Hydrostatic boundary pressures)**.** *At the inlet (submerged): $p_{in} = p_{atm} + \rho g(h_c - h_0)$. At the front (open to air): $p_{front} = p_{atm}$.*

The pressure difference driving the flow is:

$$\Delta p = p_{\text{in}} - p_{\text{front}} = \rho g(h_c - h_0). \tag{15}$$

Combined with the gravity term $\rho g(h_0 - h(L))$ acting on the slug:

$$\text{Net driving head} = (h_c - h_0) + (h_0 - h(L)) = h_c - h(L). \tag{16}$$

The inlet elevation h_0 cancels—only the difference between container level and front elevation matters.

What this neglects: Dynamic pressure at boundaries, flow contraction at inlet.

3.3 The Governing Equation

Combining all assumptions, the momentum equation for the filling slug becomes:

$$\boxed{L\ddot{L} = g[h_c - h(L)] - \left(\frac{fL}{D} + \sum K\right)\frac{\dot{L}^2}{2}} \tag{17}$$

where we have included minor losses $\sum K$ at bends and fittings, following standard practice (Idelchik, 1994).

The left-hand side represents inertia: as the slug grows (mass $\propto L$) while accelerating, Newton's second law gives $F = ma = (\rho AL)\ddot{L}$. The right-hand side contains the driving head (first term) and friction losses (second term).

3.4 Summary of Approximations

Assumption	Mathematical Effect	Physical Neglect
Axisymmetric flow	3D $\to$ 2D	Bends, non-circular pipes
Slender pipe	$\partial_s \ll \partial_r$	Short pipes
Quasi-steady profile	Algebraic τ_w	Unsteady friction
Cross-sectional averaging	ODE in s only	Radial variations
Rigid slug	ODE in t only	Compressibility
Hydrostatic boundaries	Algebraic Δp	Dynamic inlet effects

4 The Frictionless Model

We begin with the simplest case: no friction and constant container level. This isolates the role of geometry.

4.1 Equation of Motion

Setting $f = 0$, $\sum K = 0$, and $h_c = h_c^{(0)}$ in (17):

$$L\ddot{L} = g[h_c^{(0)} - h(L)]. \tag{18}$$

4.2 The Driving Force

The net force on the slug is:

$$F = \rho g A[h_c^{(0)} - h(L)]. \tag{19}$$

Physical interpretation:

- When $h(L) < h_c^{(0)}$: Front below waterline $\Rightarrow F > 0$ (acceleration)
- When $h(L) > h_c^{(0)}$: Front above waterline $\Rightarrow F < 0$ (deceleration)
- When $h(L) = h_c^{(0)}$: Front at waterline $\Rightarrow F = 0$ (coasting)

4.3 Variable-Mass Dynamics

The acceleration is:

$$a = \frac{F}{\rho AL} = \frac{g[h_c^{(0)} - h(L)]}{L}. \tag{20}$$

The $1/L$ factor is crucial: even when the force is large (e.g., at low point C where $h_c^{(0)} - h(C) = 7.5\,\text{cm}$), the acceleration is suppressed by the growing slug length. This is characteristic of variable-mass systems (Plastino & Muzzio, 1992).

4.4 The Energy Integral

Using the identity $\ddot{L} = v\,dv/dL$ where $v = \dot{L}$:

$$v\frac{dv}{dL} = \frac{g[h_c^{(0)} - h(L)]}{L}. \tag{21}$$

Integrating from $L = S_1$ (where $v = 0$, node B) to position L:

$$\frac{v^2}{2} = g\int_{S_1}^{L} \frac{h_c^{(0)} - h(r)}{r}\,dr. \tag{22}$$

Definition 1 (Energy Integral)**.** The *frictionless energy integral* is:

$$\boxed{I(L) = \int_{S_1}^{L} \frac{h_c^{(0)} - h(r)}{r}\,dr} \tag{23}$$

The velocity is then $v(L) = \sqrt{2gI(L)}$ for $I(L) > 0$.

4.5 Physical Interpretation

The integrand has two factors:

1. $h_c^{(0)} - h(r)$: The driving head at position r. Positive below waterline, negative above.
2. $1/r$: Variable-mass weighting. Early contributions (short slug) receive greater weight per unit length.

This weighting has important implications:

- **Steep early drops are most effective.** The $1/r$ factor means the drop $B \to C$ contributes more than an equivalent drop later in the path.
- **Energy stored early matters more.** The integral accumulates "weighted energy" rather than simple energy.

4.6 Evaluation for Piecewise-Linear Geometry

On a segment where $h(r) = c + sr$, the contribution from interval $[a, b]$ is:

$$\Phi(a, b; c, s) = \int_a^b \frac{h_c^{(0)} - c - sr}{r}\,dr = (h_c^{(0)} - c)\ln\frac{b}{a} - s(b - a). \tag{24}$$

The total integral is computed by summing contributions from each segment.

4.7 Priming Criterion

Priming Criterion

Priming succeeds in the frictionless limit if and only if:

$$I(S_W) > 0. \tag{25}$$

Beyond point W, the front remains below the waterline and $I(L)$ only increases. Any geometry failing this criterion cannot prime, regardless of friction reduction.

4.8 Worked Example

For our experimental geometry:

$$\Phi_2 = (12.5 - 19.84)\ln\frac{20.99}{12.50} - (-0.707)(8.49) = +2.20\,\text{cm}, \tag{26}$$

$$\Phi_3 = (12.5 + 9.93)\ln\frac{32.23}{20.99} - (0.712)(11.24) = +1.62\,\text{cm}, \tag{27}$$

$$\Phi_4 = (12.5 - 40.57)\ln\frac{32.81}{32.23} - (-0.855)(0.58) \approx 0\,\text{cm}. \tag{28}$$

Total: $I(S_W) = 3.82\,\text{cm} > 0$ (priming succeeds).
Predicted velocity at W: $v_W = \sqrt{2 \times 981 \times 3.82} = 86.5\,\text{cm/s}$.

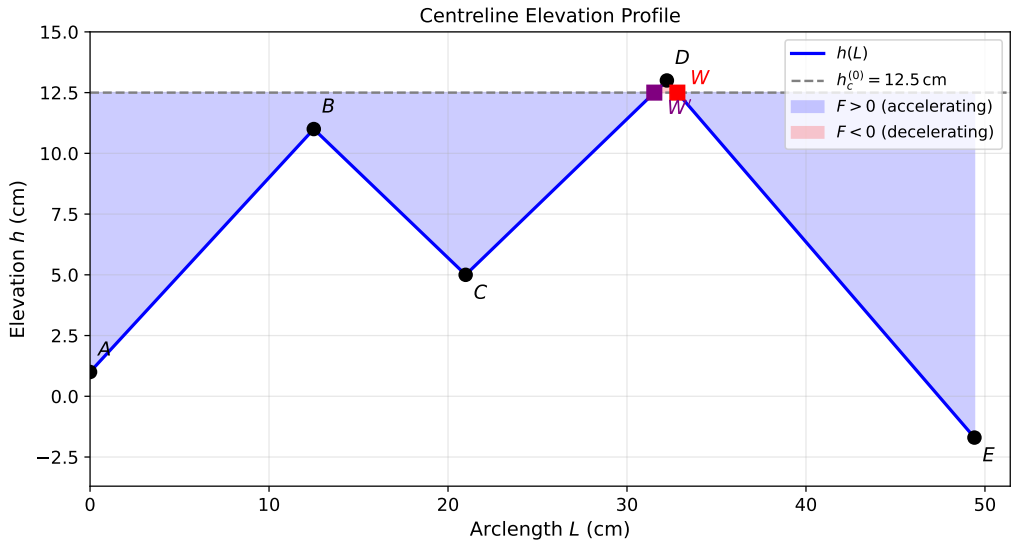


Figure 2: Centreline elevation $h(L)$ versus arclength. The dashed line marks $h_c^{(0)} = 12.5\,\text{cm}$. Blue shading indicates positive driving force (front below waterline); red shading indicates negative driving force.

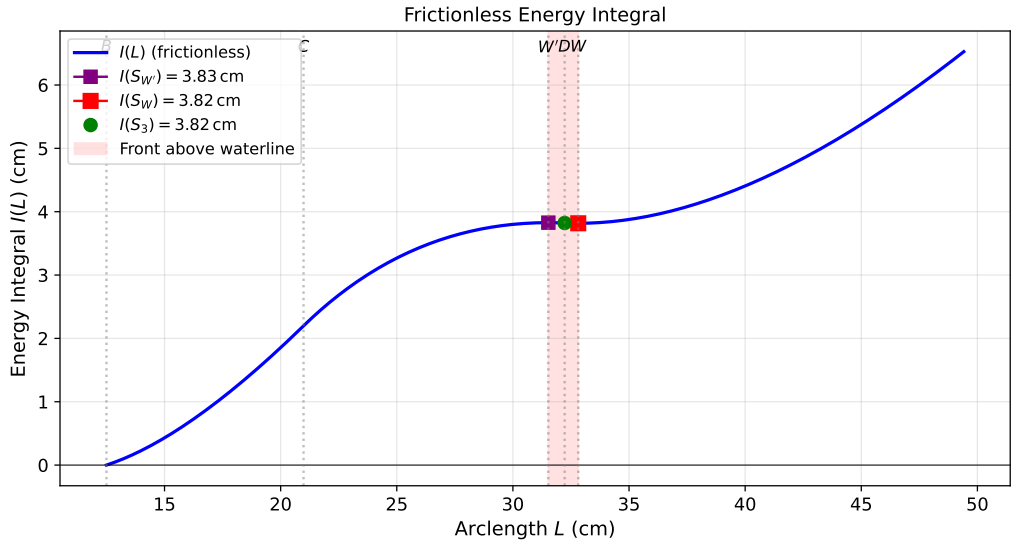


Figure 3: Energy integral $I(L)$ versus arclength. The integral rises during the drop, continues rising while the front remains below the waterline on the climb, then decreases slightly between W' and W. At S_W, $I(S_W) = 3.82\,\text{cm} > 0$: the frictionless priming criterion is satisfied.

5 Adding Container Drawdown

As water enters the pipe, the container level drops. This *drawdown* reduces the driving force.

5.1 Volume Conservation

Water entering the pipe equals water leaving the container:

$$A(L - S_1) = A_0[h_c^{(0)} - h_c(L)]. \tag{29}$$

Solving for the container level:

$$\boxed{h_c(L) = h_c^{(0)} - \frac{A}{A_0}(L - S_1)} \tag{30}$$

The level decreases linearly with slug position at rate A/A_0.

5.2 Modified Energy Integral

With variable container level, the energy integral becomes:

$$I_{\text{dd}}(L) = \int_{S_1}^{L} \frac{h_c(r) - h(r)}{r}\, dr = I(L) - \frac{A}{A_0}\int_{S_1}^{L} \frac{r - S_1}{r}\, dr. \tag{31}$$

The drawdown penalty integral evaluates to:

$$\int_{S_1}^{L} \frac{r - S_1}{r}\, dr = (L - S_1) - S_1 \ln \frac{L}{S_1}. \tag{32}$$

5.3 When Drawdown Matters

The total drawdown by the outlet:

$$\Delta h_c = \frac{A}{A_0}(L_{\max} - S_1). \tag{33}$$

For our experiment: $\Delta h_c = (5.77 \times 10^{-4}) \times 36.91 = 0.021\,\text{cm}$—negligible.

For a smaller container with $A_0 = 20\,\text{cm}^2$: $\Delta h_c \approx 0.36\,\text{cm}$, which is 24% of the initial driving head at node B.

Key Result

The area ratio A/A_0 controls drawdown importance:

- $A/A_0 \ll 0.01$: Drawdown negligible
- $A/A_0 \gtrsim 0.01$: Drawdown must be included

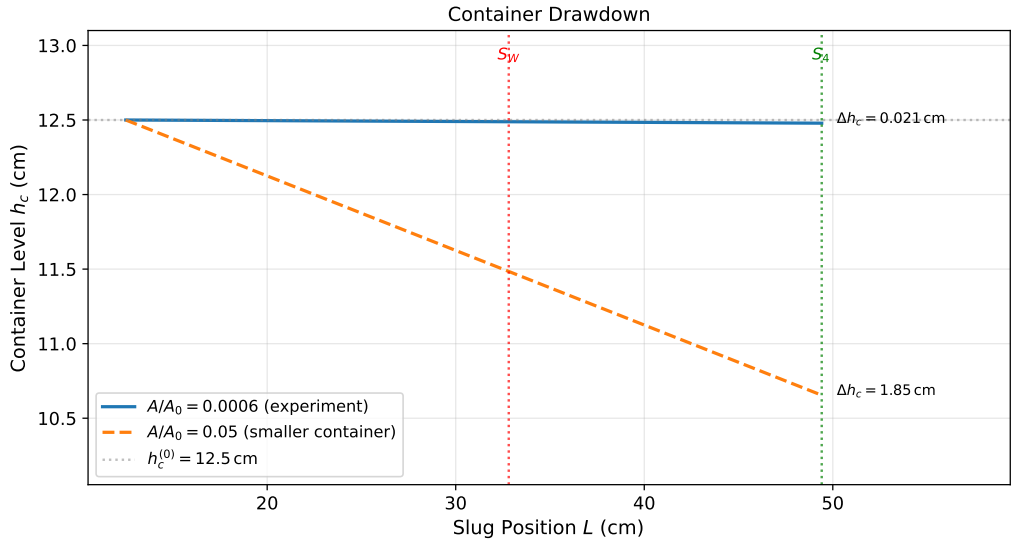


Figure 4: Container level $h_c(L)$ for two area ratios. The experimental case (solid) shows negligible drawdown; a smaller container (dashed) shows significant reduction in driving head.

6 Adding Friction

Viscous friction dissipates kinetic energy. We develop analytical results using constant friction, then discuss velocity-dependent friction.

6.1 The Darcy-Weisbach Equation

The head loss over pipe length L is given by the Darcy-Weisbach equation (Brown, 2002; Darcy, 1857):

$$h_f = f\frac{L}{D}\frac{v^2}{2g}, \tag{34}$$

where f is the Darcy friction factor. This formulation, developed by Darcy and Weisbach in the 1840s and standardized by Moody (1944), has become the universal method for pipe friction calculations.

Minor losses at bends and fittings add (Idelchik, 1994):

$$h_{\text{minor}} = \sum K \frac{v^2}{2g}, \tag{35}$$

where K values are tabulated for various fittings. Typical values range from $K \approx 0.3$ for gentle bends to $K \approx 1.5$ for sharp 90° elbows.

6.2 The Friction Factor

The friction factor depends on Reynolds number $\text{Re} = vD/\nu$ and relative roughness ε/D. In the laminar regime ($\text{Re} < 2300$), the Hagen-Poiseuille solution gives (Hagen, 1839; Poiseuille, 1840):

$$f = \frac{64}{\text{Re}} \quad \text{(laminar)}. \tag{36}$$

In the turbulent regime, the Colebrook-White equation (Colebrook, 1939) provides:

$$\frac{1}{\sqrt{f}} = -2\log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\text{Re}\sqrt{f}}\right) \quad \text{(turbulent)}. \tag{37}$$

For calculations spanning all regimes, we use the Churchill correlation (Churchill, 1977):

$$f = 8\left[\left(\frac{8}{\text{Re}}\right)^{12} + \frac{1}{(A_C + B_C)^{3/2}}\right]^{1/12}, \tag{38}$$

where A_C and B_C are functions of Re and ε/D given in appendix A. This correlation recovers the Hagen-Poiseuille result for laminar flow and matches Colebrook-White for turbulent flow.

6.3 Constant-Friction Approximation

Treating f as constant enables analytical solution. With friction, the equation of motion becomes:

$$v\frac{dv}{dL} + \left(\frac{f_0}{D} + \frac{\sum K}{L}\right)\frac{v^2}{2} = \frac{g[h_c^{(0)} - h(L)]}{L}. \tag{39}$$

Substituting $u = v^2/2$ gives a linear first-order ODE:

$$\frac{du}{dL} + \left(\frac{f_0}{D} + \frac{\sum K}{L}\right)u = \frac{g[h_c^{(0)} - h(L)]}{L}. \tag{40}$$

The integrating factor is:

$$\mu(L) = \left(\frac{L}{S_1}\right)^{\sum K} \exp\left(\frac{f_0(L - S_1)}{D}\right). \tag{41}$$

Definition 2 (Friction-Corrected Energy Integral)**.**

$$\boxed{I_f(L) = \frac{1}{\mu(L)}\int_{S_1}^{L} \mu(r)\frac{h_c^{(0)} - h(r)}{r}\,dr} \tag{42}$$

6.4 Physical Interpretation

The effective weight for energy stored at position r becomes:

$$\frac{\mu(r)}{\mu(L)} \cdot \frac{1}{r} = \frac{1}{r}\left(\frac{r}{L}\right)^{\sum K} \exp\left(-\frac{f_0(L - r)}{D}\right). \tag{43}$$

The exponential factor *discounts* early contributions: energy stored early must travel farther and suffers more dissipation. This explains why $I_f(L) < I(L)$ always.

6.5 Loss Estimation

For our experiment, the measured peak velocity is $v \approx 36\,\text{cm/s}$, giving:

$$\text{Re} = \frac{36 \times 0.5}{0.01} \approx 1800 \quad \text{(mostly laminar)}. \tag{44}$$

From (36), $f \approx 64/1800 \approx 0.036$.
Distributed friction loss over the full pipe:

$$h_f \approx 0.036 \times \frac{49.4}{0.5} \times \frac{36^2}{2 \times 981} \approx 2.4\,\text{cm}. \tag{45}$$

Minor losses with fitted $\sum K \approx 14.4$:

$$h_{\text{minor}} \approx 14.4 \times \frac{36^2}{2 \times 981} \approx 9.5\,\text{cm}. \tag{46}$$

Minor losses dominate the energy budget for small-diameter siphons with tight bends.

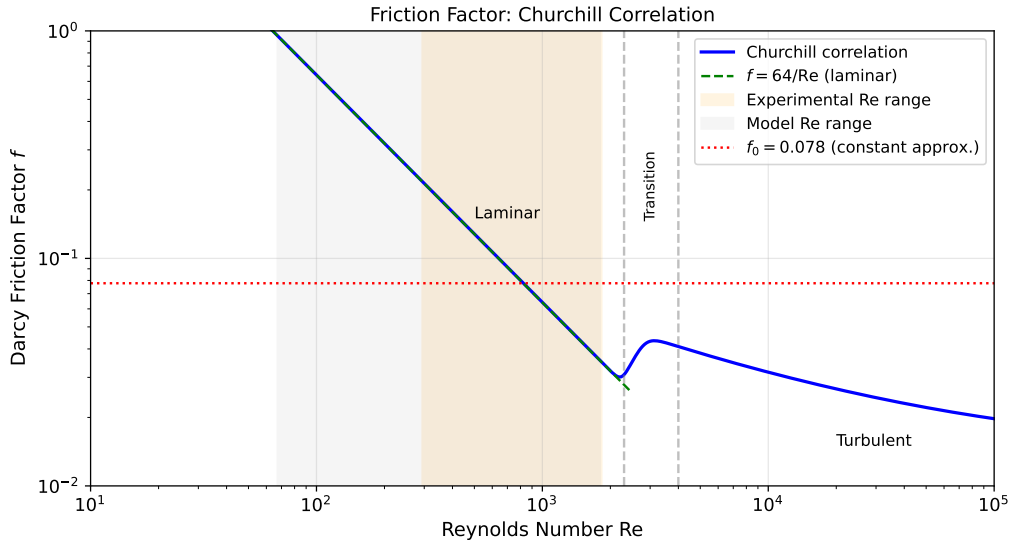


Figure 5: Friction factor f versus Reynolds number from the Churchill correlation. The experimental range (shaded) is mostly in the laminar regime. The dashed line shows the constant approximation f_0.

7 Comparing the Three Models

The three energy integrals form a hierarchy:

$$I(L) \geq I_{\text{dd}}(L) \geq I_f(L). \tag{47}$$

7.1 Values at Critical Point S_W

Model	Integral at S_W	Velocity at W	Controlling Parameter
Frictionless	$I = 3.82\,\text{cm}$	$86.5\,\text{cm/s}$	Geometry
With drawdown	$I_{\text{dd}} = 3.81\,\text{cm}$	$86.4\,\text{cm/s}$	A/A_0
With friction	$I_f \approx 0.02\,\text{cm}$	$6.3\,\text{cm/s}$	ε/D, $\sum K$

7.2 Physical Insights

- **Frictionless integral:** The upper bound. Represents maximum kinetic energy from geometry alone.
- **Drawdown correction:** Negligible for large containers (our case).
- **Friction correction:** Dominates the energy budget. The 99% reduction reflects severe discounting of early-stored energy by the $(L/S_1)^{\sum K}$ factor.

The constant-friction model is *conservative*: it provides a lower bound that, while pessimistic, guarantees priming if $I_f(S_W) > 0$.

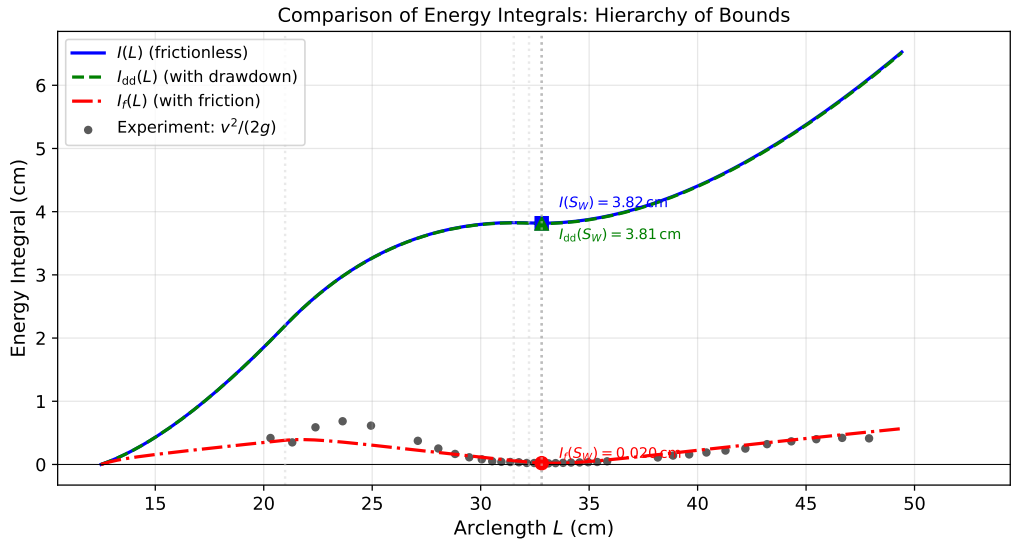


Figure 6: Comparison of energy integrals versus arclength. The frictionless integral (solid) provides the upper bound; drawdown (dashed) reduces it slightly; friction (dash-dot) causes substantial reduction. Experimental kinetic head $v^2/(2g)$ shown as points.

8 Full Numerical Dynamics

The analytical integrals provide bounds, but accurate predictions require velocity-dependent friction.

8.1 The Complete ODE System

For filling ($L < L_{\max}$):

$$\frac{dL}{dt} = v, \tag{48}$$

$$\frac{dv}{dt} = \frac{g}{L}\left[h_c(L) - h(L)\right] - \left(\frac{f(\mathrm{Re})}{D} + \frac{\sum K}{L}\right)\frac{v^2}{2}. \tag{49}$$

At each step, $\mathrm{Re} = vD/\nu$ is computed and f evaluated from (38).

8.2 Draining Phase

Once the pipe fills completely ($L = L_{\max}$), the slug length is constant:

$$\frac{dh_c}{dt} = -\frac{A}{A_0} v, \tag{50}$$

$$\frac{dv}{dt} = \frac{g[h_c - h_4]}{L_{\max}} - \left(\frac{f}{D} + \frac{\sum K}{L_{\max}} \right) \frac{v^2}{2}. \tag{51}$$

For quasi-steady draining (large containers), setting $dv/dt = 0$:

$$v_{\text{qs}} = \sqrt{\frac{2g(h_c - h_4)}{f L_{\max}/D + \sum K}}. \tag{52}$$

8.3 Draining Time Estimate

Integrating the quasi-steady approximation:

$$t_{\text{drain}} = \frac{2A_0}{A} \sqrt{\frac{f L_{\max}/D + \sum K}{2g}} \sqrt{h_c^{(0)} - h_4}. \tag{53}$$

For our experiment: $t_{\text{drain}} \approx 23\,\text{min}$—much longer than filling due to friction-limited steady-state velocity.

9 Experimental Validation

9.1 Setup

The experimental siphon was constructed from 5 mm inner diameter flexible tubing bent around a rigid frame to maintain the five-node geometry. The container was a rectangular tank with $A_0 = 340\,\text{cm}^2$ cross-section.

9.2 Video Tracking

High-speed video (120 fps) captured the filling process. The water front position was extracted frame-by-frame using edge detection. Velocity was computed from position differences using a Savitzky-Golay filter to reduce noise.

9.3 Results

Figures 7 and 8 compare model predictions with experimental data. Key observations:

- The frictionless model overestimates velocity by a factor of ~ 2.
- The constant-friction model underestimates velocity (conservative bound).
- The full numerical model captures the essential dynamics.
- Peak measured velocity: $\sim 36\,\text{cm/s}$ at low point C.
- Filling time to outlet: $\sim 2\,\text{s}$.

The effective minor loss coefficient $\sum K \approx 14.4$ was obtained by fitting the numerical model to the measured $v(L)$ curve. This value is higher than handbook estimates for smooth bends, likely due to:

1. Tight bend radii in our geometry
2. Entrance and exit effects not separately modeled
3. Developing flow in short segments

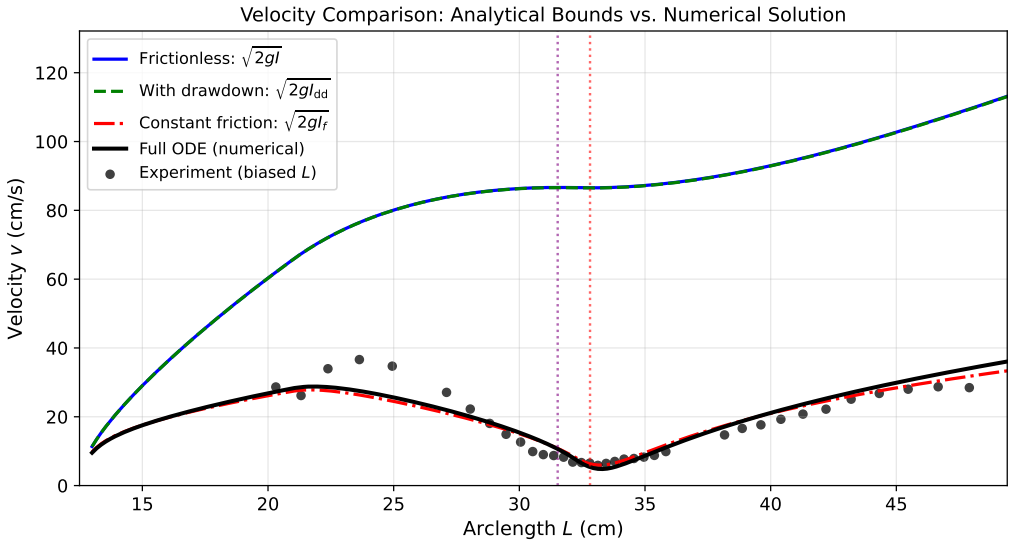


Figure 7: Velocity $v(L)$ during filling. Model curves: frictionless (upper bound), constant friction (lower bound), full ODE (best estimate). Experimental points track the full model reasonably well.

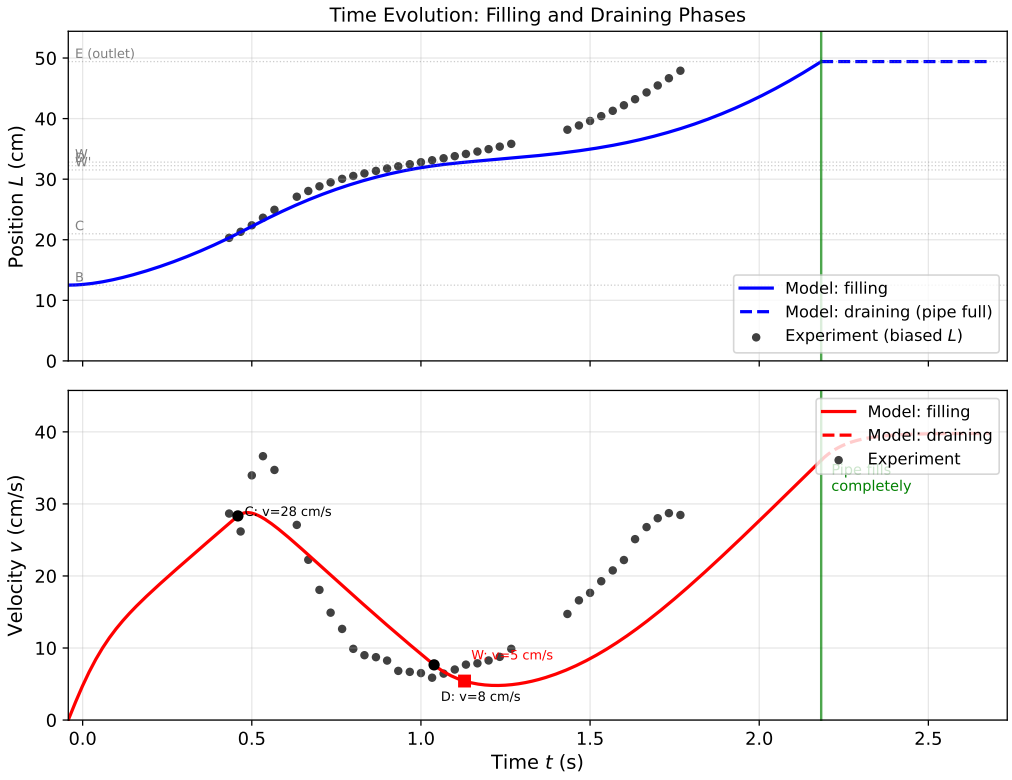


Figure 8: Position $L(t)$ and velocity $v(t)$ versus time. Model curves are time-shifted to align with experimental start. Key positions (C, D, W, outlet) are marked.

10 Validity Limits

Every model has a domain of validity. We quantify when each assumption fails using dimensionless parameters.

10.1 Cavitation

At the crest, pressure is minimum. Cavitation occurs when pressure drops below the vapor pressure $p_v \approx 2.3\,\text{kPa}$ (Brennen, 1995):

$$p_D = p_{\text{atm}} - \rho g(h_3 - h_c) - \frac{\rho v^2}{2} - \rho g h_f > p_v. \tag{54}$$

The maximum siphon height is approximately 10 m for water at sea level. For our geometry with $h_3 - h_c = 0.5\,\text{cm}$, cavitation is not a concern.

10.2 Capillarity

Surface tension effects become important at small diameters. Two dimensionless numbers characterize the regime (de Gennes et al., 2004):

Definition 3 (Bond Number)**.**

$$\text{Bo} = \frac{\rho g D^2}{\gamma} \tag{55}$$

compares gravitational to surface tension forces. Gravity dominates when $\text{Bo} \gg 1$.

Definition 4 (Capillary Number)**.**

$$\text{Ca} = \frac{\mu v}{\gamma} \tag{56}$$

compares viscous to surface tension forces. Dynamic meniscus effects are negligible when $\text{Ca} \ll 1$.

For our experiment: $\text{Bo} \approx 3.4$ and $\text{Ca} \approx 0.005$. With Bo only marginally above 1, capillary effects may contribute at the 10-20% level.

10.3 Entrance Effects

The laminar entrance length scales as (White, 2011):

$$\frac{L_e}{D} \approx 0.05\,\text{Re}. \tag{57}$$

At $\text{Re} \approx 1800$, this gives $L_e \approx 45\,\text{cm}$—comparable to segment lengths. The quasi-steady friction assumption is therefore marginal.

10.4 Secondary Flows

Flow through bends generates Dean vortices, characterized by the Dean number (Dean, 1927):

$$\text{De} = \text{Re}\sqrt{\frac{D}{2R}}, \tag{58}$$

where R is the bend radius. Secondary flows become significant when $\text{De} > 100$ (Berger et al., 1983).

For our tight bends with $R \approx 2D$: $\text{De} \approx 900$, indicating strong secondary flows that justify treating bend losses as lumped K values rather than distributed friction.

10.5 Unsteady Friction

The Womersley number characterizes oscillatory effects (Womersley, 1955):

$$\text{Wo} = D\sqrt{\frac{\omega}{\nu}}, \tag{59}$$

where $\omega \sim 1/T$ is the frequency of velocity variation. For $T \sim 0.5\,\text{s}$: $\text{Wo} \approx 5$. Unsteady effects are modest but not negligible.

10.6 Regime Diagrams

Figure 9 summarizes validity limits in three parameter spaces.

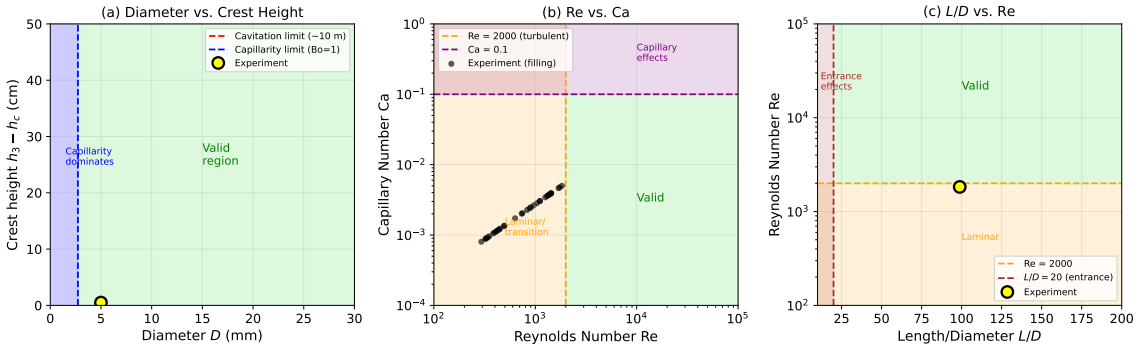


Figure 9: Validity regime diagrams. (a) Diameter vs. crest height: cavitation limits tall crests; capillarity limits small diameters. (b) Re vs. Ca: shows laminar/turbulent boundary and capillary regime. (c) L/D vs. Re: entrance effects become important at low L/D or high Re. The experimental point (marked) lies near the capillary and entrance-effect boundaries.

11 Conclusion

We developed a theoretical framework for self-starting siphons through systematic derivation from the Navier-Stokes equations and layered incorporation of physical effects.

11.1 Key Results

1. **Geometry is fundamental.** The frictionless energy integral $I(L)$ determines whether priming is geometrically possible. The criterion $I(S_W) > 0$ provides a closed-form screening tool.
2. **Friction dominates losses.** For small-diameter siphons with tight bends, minor losses ($\sum K$) typically exceed distributed friction (fL/D). Reducing bend losses is more effective than smoothing pipe walls.
3. **Drawdown matters for small containers.** When $A/A_0 > 0.01$, container level drop significantly reduces driving head.
4. **Analytical bounds bracket reality.** The hierarchy $I \geq I_{\text{dd}} \geq I_f$ provides physical insight that numerical simulation alone cannot offer.
5. **Validity limits are quantifiable.** Dimensionless parameters (Bo, Ca, De, Wo) identify when assumptions fail. Our experiment lies near the capillary and entrance-effect boundaries.

11.2 Design Guidelines

For reliable self-starting:

- Maximize the early drop ($B \to C$) to build momentum when the slug is short
- Minimize crest height ($h_3 - h_c^{(0)}$) while maintaining positive margin
- Use gentle bends with large radii to reduce K values
- Prefer larger diameters to reduce friction and capillary effects
- Ensure adequate container size if drawdown matters

11.3 Future Work

Extensions could address:

- Systematic optimization of geometry for minimum filling time
- Inclusion of capillary effects for sub-5 mm diameters
- Unsteady friction corrections for rapid transients
- Two-phase flow effects during initial priming

A Friction Correlations

A.1 The Churchill Correlation

The Churchill correlation (Churchill, 1977) provides a single formula valid across all flow regimes:

$$f = 8\left[\left(\frac{8}{\mathrm{Re}}\right)^{12} + \frac{1}{(A_C + B_C)^{3/2}}\right]^{1/12}, \tag{60}$$

where

$$A_C = \left[2.457 \ln\left(\frac{1}{(7/\mathrm{Re})^{0.9} + 0.27(\varepsilon/D)}\right)\right]^{16}, \tag{61}$$

$$B_C = \left(\frac{37530}{\mathrm{Re}}\right)^{16}. \tag{62}$$

A.2 Limiting Behavior

Laminar limit ($\mathrm{Re} \to 0$): The first term dominates: $f \to 64/\mathrm{Re}$, recovering Hagen-Poiseuille.

Turbulent limit ($\mathrm{Re} \to \infty$): The second term dominates, recovering Colebrook-White.

Transition: Smooth interpolation provides continuous, differentiable $f(\mathrm{Re})$.

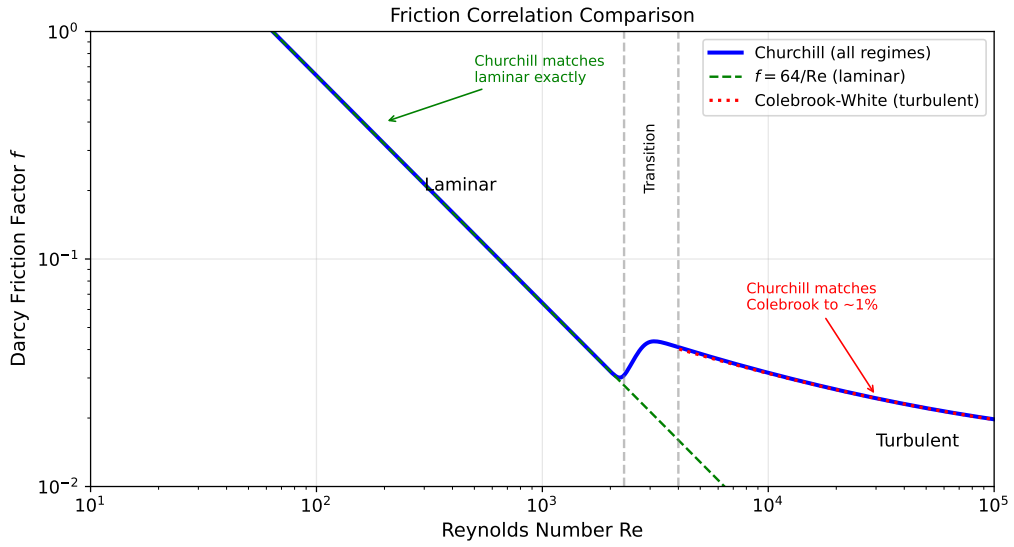


Figure 10: Comparison of friction correlations. The Churchill correlation (solid) matches the laminar law exactly and Colebrook-White (dotted, turbulent only) to within 1%.

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